

4. Project Planning

Project Name

ComicCraft - AI Comic Story Creator using Gemini Models

1. Introduction

Project planning defines the tasks, development stages, resources, and timeline required to successfully develop the ComicCraft application.

The project is divided into different phases to make development simple and organized.

2. Project Goals

- Develop an AI-powered comic story creator.
- Integrate Google Gemini AI.
- Create a simple and user-friendly web interface.
- Generate creative stories, characters, scenes, and dialogues.
- Provide a comic preview.
- Test and document the completed application.

3. Development Phases

Phase 1 – Brainstorming and Ideation

Activities:

- Identify the project idea.
- Identify the problem.
- Define the proposed solution.
- Finalize the project name.

Output: Project idea and concept.

Phase 2 – Requirement Analysis

Activities:

- Identify functional requirements.
- Identify non-functional requirements.
- Identify hardware and software requirements.
- Define user requirements.

Output: Requirement Analysis document.

Phase 3 – Project Design

Activities:

- Design system architecture.
- Design application modules.
- Design user interface.
- Define data flow.
- Plan project folder structure.

Output: Project Design document.

Phase 4 – Development

Activities:

- Create frontend pages.
- Develop Flask backend.
- Connect Gemini AI API.
- Implement story generation.
- Implement comic preview.
- Connect frontend and backend.

Output: Working ComicCraft application.

Phase 5 – Testing

Activities:

- Test user input.
- Test story generation.
- Test API communication.
- Test page navigation.
- Fix errors and bugs.

Output: Tested application.

Phase 6 – Documentation

Activities:

- Prepare README files.
- Document project features.
- Document technologies used.
- Add screenshots where required.
- Prepare project presentation.

Output: Complete project documentation.

4. Project Timeline

Phase| Task| Status 1| Brainstorming and Ideation| Completed 2| Requirement Analysis| Completed 3| Project Design| Completed 4| Development| In Progress 5| Testing| Planned 6| Documentation| Planned

5. Resources Required

Hardware

- Laptop or computer
- Minimum 4 GB RAM
- Internet connection

Software

- Visual Studio Code
- Python
- Flask
- Google Gemini API
- Google Chrome
- Git
- GitHub

6. Team Responsibilities

Role| Responsibility Project Lead| Manage overall project Frontend Developer| Design web pages Backend Developer| Develop Flask backend AI Integration| Integrate Gemini AI Tester| Test and fix application issues Documentation| Prepare project documentation

7. Risk Management

Risk| Solution Gemini API errors| Check API configuration and error handling Internet problems| Retry the request when connection is available Application bugs| Perform regular testing Invalid user input| Add input validation Development delays| Divide work into smaller tasks

8. Project Deliverables

The project will include:

- Working ComicCraft web application
- Frontend files
- Flask backend
- Gemini AI integration
- Comic preview
- Project documentation
- GitHub repository
- Project presentation

9. Success Criteria

The project will be considered successful when:

- The application accepts a user's story idea.
- Gemini AI successfully generates story content.
- Characters, scenes, and dialogues are generated.
- The generated content is displayed correctly.
- The application works without major errors.
- The project documentation is completed.

10. Conclusion

Project planning provides a clear roadmap for developing ComicCraft. Dividing the project into different phases helps organize the development process, manage time, reduce risks, and achieve the expected project outcome.